Chap 1. 复变函数.

1. 起源.

$x^3+ax^2+bx+c=0$. $x=t-\frac{a}{3}$

$t^3+pt+q=0$

$t=u+v$ $u^3+v^3+(u+v)(3uv+p)+q=0$

多出一个自由度，可任加一个条件

令 $3uv+p=0$

$$\begin{cases} u^3+v^3+q=0 \\ uv=-\frac{p}{3} \end{cases} \quad u^6+u^3v^3+qu^3=0. \quad u^6-\frac{p^3}{27}+qu^3=0.$$

$u^6+qu^3-\frac{p^3}{27}=0$

令 $y=u^3$ $y^2+qy-\frac{p^3}{27}=0$.

$$y=\frac{-q\pm\sqrt{q^2+\frac{4p^3}{27}}}{2}$$

$$u=\sqrt[3]{\frac{-q+\sqrt{q^2+\frac{4p^3}{27}}}{2}}$$

$$v=\sqrt[3]{\frac{-q-\sqrt{q^2+\frac{4p^3}{27}}}{2}}$$

$q^2+\frac{4p^3}{27}<0$.

极坐标下的柯西-黎曼条件

$\frac{\partial u}{\partial x}=\frac{\partial u}{\partial\rho}\frac{\partial\rho}{\partial x}+\frac{\partial u}{\partial\varphi}\frac{\partial\varphi}{\partial x}$

$\rho=(x^2+y^2)^{\frac{1}{2}}$

$\frac{\partial\rho}{\partial x}=\frac{x}{\rho}=\cos\varphi$

$$\frac{df(z)}{dz}=\lim_{\substack{\Delta\rho\to0\\ \Delta\varphi=0}}\frac{u(\rho+\Delta\rho,\varphi)-u(\rho,\varphi)+iv(\rho+\Delta\rho,\varphi)-iv(\rho,\varphi)}{\Delta\rho e^{i\varphi}}$$

$=\frac{\partial u}{\partial\rho}e^{-i\varphi}+i\frac{\partial v}{\partial\rho}e^{-i\varphi}$

$$=\lim_{\Delta z\to0}\frac{\Delta f(z)}{\Delta z}=\lim_{\substack{\Delta\rho=0\\ \Delta\varphi\to0}}\frac{u(\rho,\varphi+\Delta\varphi)+iv(\rho,\varphi+\Delta\varphi)-u(\rho,\varphi)-iv(\rho,\varphi)}{i\rho e^{i\varphi}\Delta\varphi}$$

$=\frac{-i}{\rho}e^{-i\varphi}\frac{\partial u}{\partial\varphi}+\frac{1}{\rho}e^{-i\varphi}\frac{\partial v}{\partial\varphi}$

$\frac{\partial u}{\partial\rho}=\frac{1}{\rho}\frac{\partial v}{\partial\varphi}$ $\frac{\partial u}{\partial\varphi}\frac{1}{\rho}=-\frac{\partial v}{\partial\rho}$

解析函数 $f(z)=u(x,y)+iv(x,y)$

定义.

① 若 $f(z)$ 在 $z=z_0$ 邻域内处处可导.

则称 f 在点 $z=z_0$ 处解析.

② 若 $f(z)$ 在区域 B 上每一点都解析.

则称 $f(z)$ 在 B 上解析

2) 性质.

① 在解析区域内等 u 线 (即 $u(x,y)=C_1$)

和等 v 线 ($v(x,y)=C_2$) 相互正交!

$\nabla u\cdot\nabla v=0$.

$\nabla u=\frac{\partial u}{\partial x}\cdot\vec{e_x}+\frac{\partial u}{\partial y}\cdot\vec{e_y}$

$\nabla v=\frac{\partial v}{\partial x}\cdot\vec{e_x}+\frac{\partial v}{\partial y}\cdot\vec{e_y}$

$\nabla u\cdot\nabla v=\frac{\partial u}{\partial x}\cdot\frac{\partial v}{\partial x}+\frac{\partial u}{\partial y}\cdot\frac{\partial v}{\partial y}$

$=\frac{\partial v}{\partial y}\cdot\frac{\partial v}{\partial x}-\frac{\partial v}{\partial x}\cdot\frac{\partial v}{\partial y}$

$=0$

② 实部 u 和虚部 v 均为调和函数

$\Delta u=0$ $\Delta v=0$. 即 $\frac{\partial^2u}{\partial x^2}+\frac{\partial^2u}{\partial y^2}=0$.

$\Delta=\frac{\partial^2}{\partial x^2}+\frac{\partial^2}{\partial y^2}=\nabla\cdot\nabla=(\vec{e_x}\frac{\partial}{\partial x}+\vec{e_y}\frac{\partial}{\partial y})\cdot(\vec{e_x}\frac{\partial}{\partial x}+\vec{e_y}\frac{\partial}{\partial y})$

$\nabla=\vec{e_x}\frac{\partial}{\partial x}+\vec{e_y}\frac{\partial}{\partial y}$

$\nabla=\vec{e_\rho}\frac{\partial}{\partial\rho}+\vec{e_\varphi}\frac{1}{\rho}\frac{\partial}{\partial\varphi}$

$\vec{e_x}\cdot\frac{\partial}{\partial x}(\vec{e_x}\frac{\partial}{\partial x})=\vec{e_x}\cdot\vec{e_x}\frac{\partial}{\partial x}\frac{\partial}{\partial x}+\underbrace{(\vec{e_x}\cdot\frac{\partial\vec{e_x}}{\partial x})}_{=0}\frac{\partial}{\partial x}$

$=\frac{\partial^2}{\partial x^2}$

$\vec{e_x}\cdot\frac{\partial}{\partial x}(\vec{e_x}\frac{\partial}{\partial x})f(x,y)$

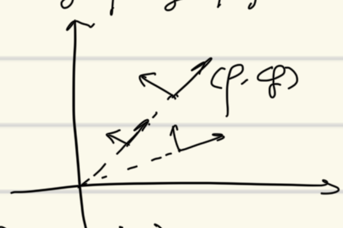
$=\vec{e_x}\cdot\frac{\partial}{\partial x}(\vec{e_x}\cdot\frac{\partial}{\partial x}f(x,y))$

$=(\vec{e_x}\cdot\frac{\partial\vec{e_x}}{\partial x}\cdot\frac{\partial}{\partial x}+\vec{e_x}\cdot\vec{e_x}\cdot\frac{\partial^2}{\partial x^2})f(x,y)$

$\nabla\cdot\nabla=(\vec{e_\rho}\frac{\partial}{\partial\rho}+\vec{e_\varphi}\cdot\frac{1}{\rho}\frac{\partial}{\partial\varphi})\cdot(\vec{e_\rho}\frac{\partial}{\partial\rho}+\vec{e_\varphi}\cdot\frac{1}{\rho}\frac{\partial}{\partial\varphi})$

$=\vec{e_\rho}\frac{\partial}{\partial\rho}(\vec{e_\rho}\frac{\partial}{\partial\rho})+\vec{e_\rho}\cdot\frac{\partial}{\partial\rho}(\vec{e_\varphi}\frac{1}{\rho}\frac{\partial}{\partial\varphi})$

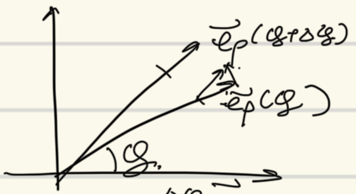
$+\vec{e_\varphi}\frac{1}{\rho}\frac{\partial}{\partial\varphi}(\vec{e_\rho}\frac{\partial}{\partial\rho})+\vec{e_\varphi}\frac{1}{\rho}\frac{\partial}{\partial\rho}(\vec{e_\rho}\frac{1}{\rho}\frac{\partial}{\partial\varphi})$



$\vec{e_\rho}(\varphi)$ $\vec{e_\varphi}(\varphi)$

① ... $=\frac{\partial^2}{\partial\rho^2}+\vec{e_\varphi}\cdot\frac{\partial\vec{e_\rho}}{\partial\varphi}\cdot\frac{1}{\rho}\frac{\partial}{\partial\rho}+\vec{e_\varphi}\cdot\frac{\partial\vec{e_\varphi}}{\partial\varphi}\cdot\frac{1}{\rho}\frac{\partial}{\partial\varphi}+\frac{1}{\rho^2}\frac{\partial^2}{\partial\varphi^2}$

$$\frac{\partial \vec{e}_\rho}{\partial \varphi} = \lim_{\Delta\varphi\to 0} \frac{\vec{e}_\rho(\varphi+\Delta\varphi)-\vec{e}_\rho(\varphi)}{\Delta\varphi}$$



$$= \lim_{\Delta\varphi\to 0} \frac{\Delta\varphi\, \vec{e}_\varphi}{\Delta\varphi} = \vec{e}_\varphi$$

$$\frac{\partial \vec{e}_\varphi}{\partial \varphi} = \lim_{\Delta\varphi\to 0} \frac{\vec{e}_\varphi(\varphi+\Delta\varphi)-\vec{e}_\varphi(\varphi)}{\Delta\varphi}$$

$$= -\vec{e}_\rho$$

$$\frac{\partial \vec{e}_\rho}{\partial \rho} = 0, \quad \frac{\partial \vec{e}_\varphi}{\partial \rho} = 0$$

$$\frac{\partial \vec{e}_\varphi}{\partial \varphi} = -\vec{e}_\rho \quad \frac{\partial \vec{e}_\rho}{\partial \varphi} = \vec{e}_\varphi$$

以①继续: $\Delta = \frac{\partial^2}{\partial \rho^2} + \frac{1}{\rho}\frac{\partial}{\partial \rho} + \frac{1}{\rho^2}\frac{\partial^2}{\partial \varphi^2}$

$$\frac{\partial u}{\partial x} = \frac{\partial v}{\partial y}$$

$$\frac{\partial u}{\partial y} = -\frac{\partial v}{\partial x}$$

$u_{xx} = v_{xy} \quad u_{yy} = -v_{yx} = -v_{xy}$

③ $f(z)$ 为解析函数.

则已知 u 可求 v.

或已知 v 可求 u.

例 如已知 $u(x,y)$

$$dv = \frac{\partial v}{\partial x}dx + \frac{\partial v}{\partial y}dy$$

$$= -\frac{\partial u}{\partial y}dx + \frac{\partial u}{\partial x}dy$$

$v(x,y) = \int dv$. 方法: ①曲线积分 ②全微分 ③不定积分

$u(x,y) = x^2 - y^2$

$dv = 2y\,dx + 2x\,dy = d(2xy)$

$v = 2xy + C$.

$$f(z) = u + iv = x^2 - y^2 + i2xy + iC.$$

$$= (x+iy)^2 + iC.$$

$$= z^2 + iC$$

$$\frac{df(z)}{dz} = 2z.$$

$v = \sqrt{-x+\sqrt{x^2+y^2}}$, 求 $f(z)$

$$v = \sqrt{-\rho\cos\varphi + \rho} = \sqrt{\rho}\sqrt{1-\cos\varphi} = \sqrt{\rho}\sqrt{2\sin^2\frac{\varphi}{2}}$$

$$= \sqrt{2\rho}\,\sin\frac{\varphi}{2}$$

$$du(\rho,\varphi) = \frac{\partial u}{\partial \rho}d\rho + \frac{\partial u}{\partial \varphi}d\varphi$$

$$\Rightarrow f(z) = \sqrt{2\rho}\,e^{i\frac{\varphi}{2}} = \sqrt{2z}$$